Data Science

Ensemble Methods

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Outline

1. Introduction
2. General principles
3. Bagging
4. Random Forests
5. Boosting

Ensemble Learning and Random Forests

If you aggregate the predictions of a group of predictors, you will often get better predictions than with the best individual predictor. A group of predictors is called an *ensemble*; thus, this technique is called Ensemble Learning.

For example, you can train a group of Decision Tree classifiers, each on a different random subset of the training set. Such an ensemble of Decision Trees is called a Random Forest.

In this section, we will discuss the most popular Ensemble methods, including bagging and boosting. We will also explore Random Forests.

Setting

Standard supervised learning

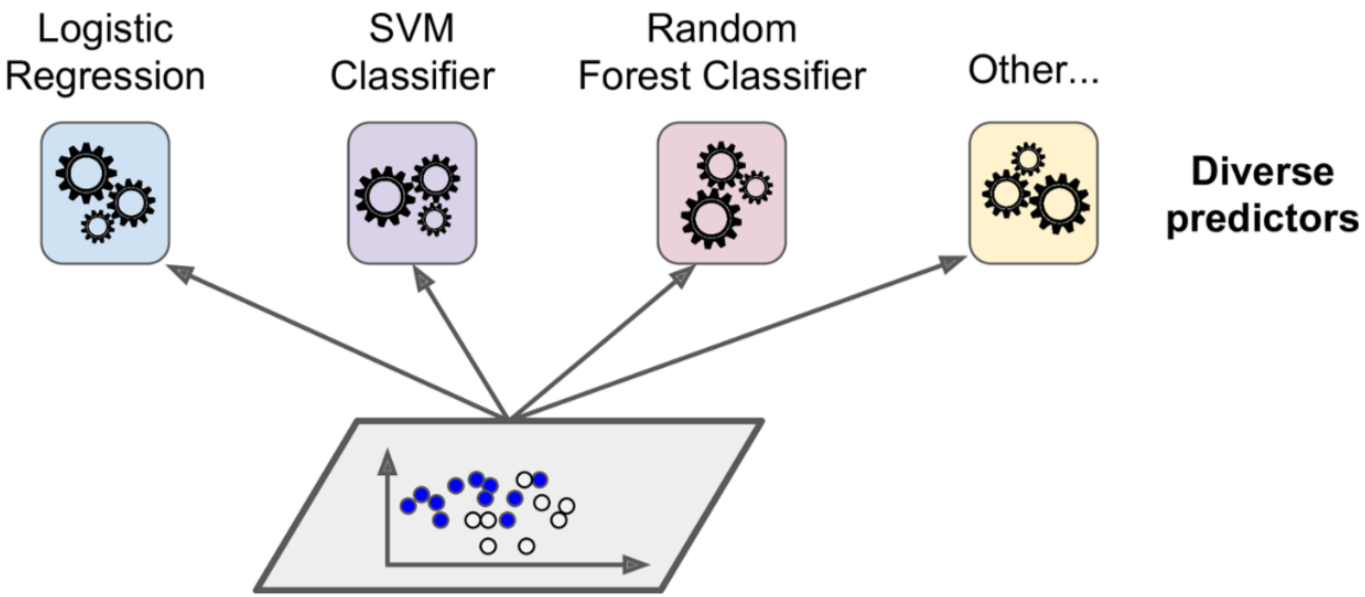
- $\mathcal{X}$ the "input" space and $\mathcal{Y}$ the "output" space
- D a fixed and unknown distribution on $\mathcal{X} \times \mathcal{Y}$
- a loss function ℓ
- a data set $\mathcal{D} = ((\mathbf{X}_i, \mathbf{Y}_i))_{1 \leq i \leq N}$ with $\mathcal{D} \sim D^N$

Combining models

- several models: $g_1, \ldots, g_K$, K functions from $\mathcal{X}$ to $\mathcal{Y}$
- can one define g from $g_1, \ldots, g_K$ in order to get better performances than with using only one of the g_k?

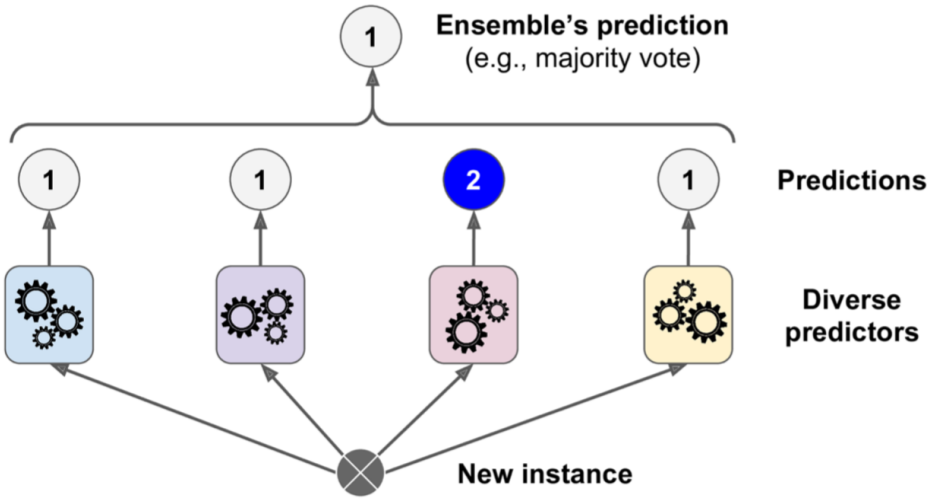
Voting Classifiers: Bagging and Pasting

- Suppose you have trained a few classifiers, each one achieving about 80% accuracy.



Voting Classifiers: Bagging and Pasting

- Create an even better classifier by aggregating the predictions of each classifier and predict the class that gets the most votes.
- This majority-vote classifier is called a hard voting classifier.
- Necessary condition: the classifiers are sufficiently diverse.

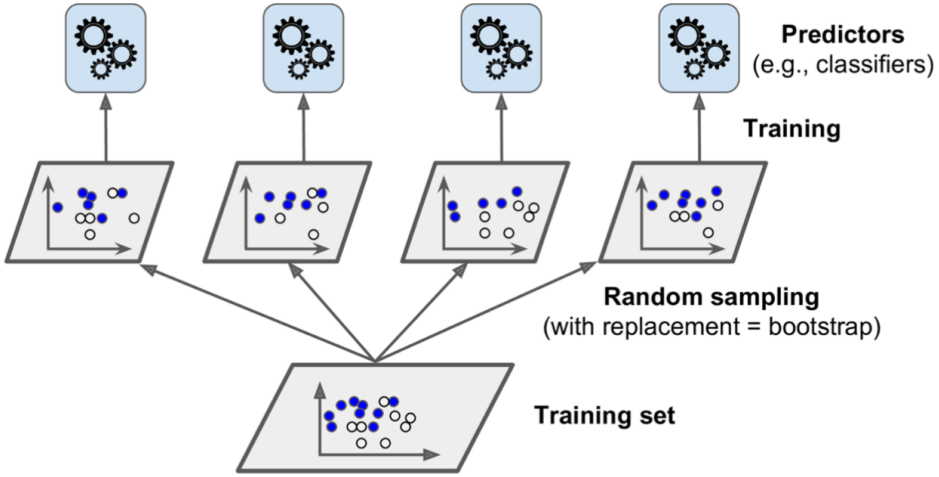


Voting Classifiers: Bagging and Pasting

- Suppose you build an ensemble containing 1,000 classifiers that are individually correct only 51% of the time.
  - Suppose these classifiers are independent.
  - If you predict the majority voted class, you can hope for up to 72.6% accuracy.
    - $X \sim \mathcal{B}(n = 1000, p = 0.51) \Rightarrow \mathbb{P}(X > 500) = 0.7260$.
- However, this is only true if all classifiers are perfectly independent, making uncorrelated errors, which is clearly not the case in case they are trained on the same data.

Voting Classifiers: Bagging and Pasting

- A common approach is to use for every predictor the same algorithm and train them on different random subsets of the training set.
  - Bagging: sampling with replacement (short for bootstrap aggregating).
  - Pasting: sampling without replacement.



Voting Classifiers: Bagging and Pasting

- Each individual predictor has a higher bias than if it were trained on the original training set, but aggregation reduces both bias and variance.
- Generally, the ensemble has a similar bias but a lower variance than a single predictor trained on the original training set.
- Also predictors can all be trained in parallel!

Majority voting

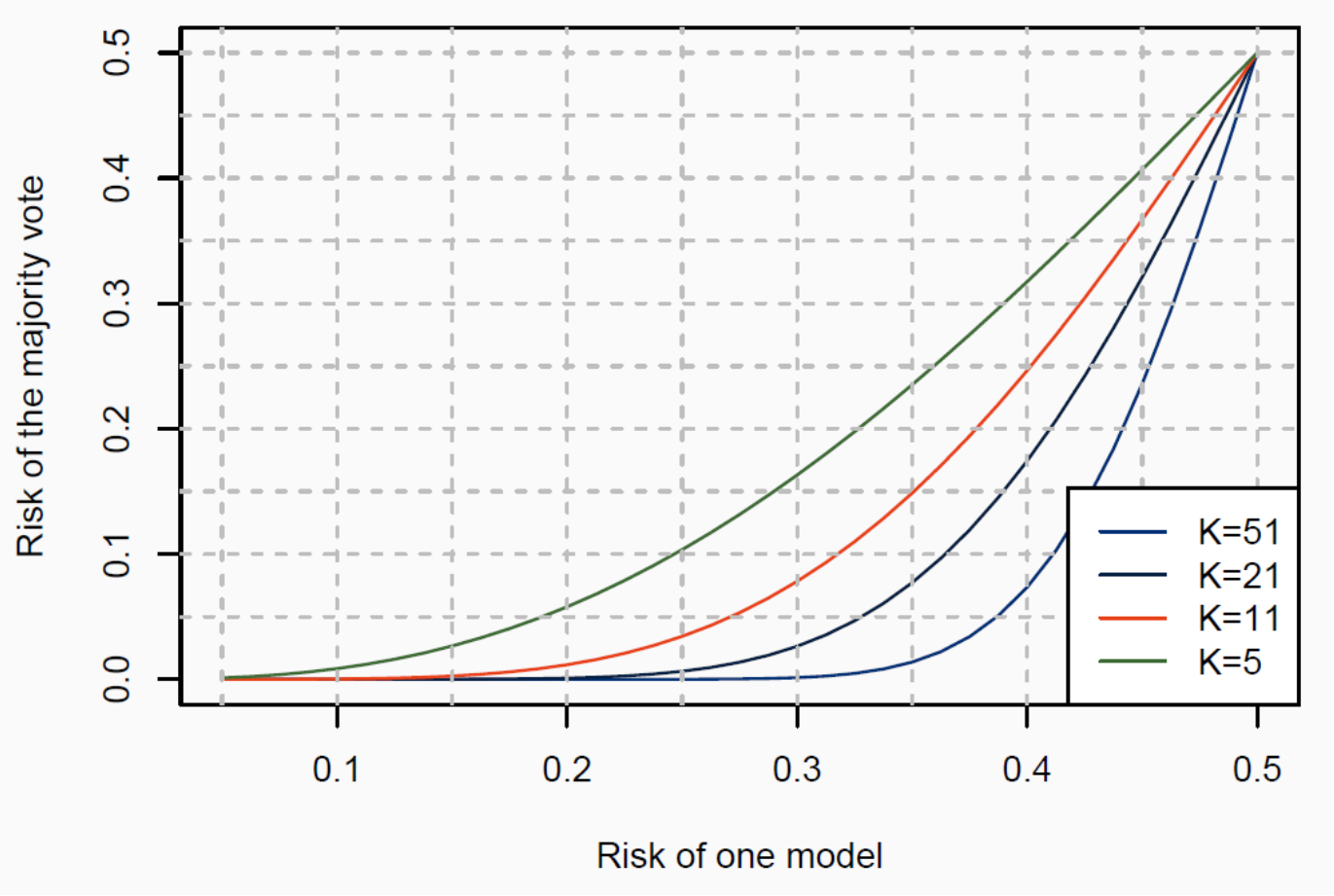
A simple example

- $\mathcal{Y} = \{-1, 1\}$ and $\ell(p, t) = \mathbb{1} p \neq t$
- $g(\mathbf{x}) = \arg\max_{y \in \mathcal{Y}} |\{1 \leq k \leq K \mid g_k(\mathbf{x}) = y\}|$ (majority voting)
- is g better than the g_k?

Analysis

- (partially unrealistic) assumptions:
  - $\forall k, R_\ell(g_k) = p < \frac{1}{2}$
  - the $g_k(\mathbf{x})$ are independent random variables (!)
- $\mathbb{P}(g(\mathbf{X}) \neq \mathbf{Y}) = \mathbb{P}\left(|\{1 \leq k \leq K \mid g_k(\mathbf{X}) \neq \mathbf{Y}\}| > \frac{K}{2}\right)$
- Binomial random variable with parameter p and K
  - $p = 0.45$ and $K = 21$: $R_\ell(g) = 0.321$
  - $p = 0.3$ and $K = 21$: $R_\ell(g) = 0.0264$

General behavior



Main ideas

To combine models

we need

- models with at least some predictive capabilities
- with independent errors

Bias variance decomposition

- another argument for model combination
- data point of view rather than model point of view:
  - a unique learning algorithm
  - several data sets
- but a similar conclusion

Ensemble methods

Techniques to combine models

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Two main questions

Building an ensemble method

1. given K models, how to combine them efficiently?
2. given a data set and some learning strategies, how to produce K models that have "independent" errors?

Combining models

Combination

- $g_1, \ldots, g_K$, K functions from $\mathcal{X}$ to $\mathcal{Y}$
- how to build g from $g_1, \ldots, g_K$?
- numerous solutions related to the nature of $\mathcal{Y}$ and to ℓ

Linear combination

- for $\mathcal{Y} = \mathbb{R}$ (or $\mathbb{R}^Q$)
- define

$$g(\mathbf{x}) = \sum_{k=1}^{K} w_k g_k(\mathbf{x})$$

- one *could* learn the $(w_k)_{1 \leq k \leq K}$ from the data but:
  - risk of overfitting
  - when there is no overfitting, the gain is small
- frequent solution $w_k = \frac{1}{K}$

Discrete predictions

Discrete case

- when $|\mathcal{Y}| < \infty$
- majority voting

$$g(\mathbf{x}) = \arg\max_{y \in \mathcal{Y}} |\{1 \leq k \leq K \mid g_k(\mathbf{x}) = y\}|$$

Binary case with score

- when $\mathcal{Y} = \{-1, 1\}$ and $g_k(\mathbf{x}) = \text{sign}(f_k(\mathbf{x}))$
- linear combination

$$g(\mathbf{x}) = \text{sign}\left(\sum_{k=1}^{K} w_k f_k(\mathbf{x})\right)$$

Probabilities and more

Probabilities

- when $|\mathcal{Y}| < \infty$
- $g_k(\mathbf{y}, \mathbf{x})$ estimates $\mathbb{P}(\mathbf{Y} = \mathbf{y} | \mathbf{X} = \mathbf{x})$
- weighted product (Z is a normalization constant)

$$g(\mathbf{y}, \mathbf{x}) = \frac{1}{Z} \prod_{k=1}^{K} g_k(\mathbf{y}, \mathbf{x})^{w_k}$$

Numerous other solutions

- mostly based on the concept of generalized mean
- adapted to special cases such as ranking (or even unsupervised settings)
- but in general, the mean, the uniform product or the majority voting are sufficient

Building models

Diversity

- is mandatory!
- most learning algorithms are deterministic: same data $\Rightarrow$ same model
- two general solutions:
  1. introduce some randomness in the algorithm (or leverage the natural randomness of the algorithm)
  2. run the same algorithm on modified data (randomly or deterministically)

Data level diversity

Instance subsets

- each classifier is obtained on a "subset" of the data set
- *Bagging*: bootstrap sample
- *Cross-validated committee*: block decomposition

Feature subsets

- each classifier is obtained using only a subset of the features
- *Random Subspace Method*: as is
- *Random Forests*: combined with *Bagging*

Data level diversity

Weighted instances

- each classifier is obtained using a weighted version of the data set
- *Boosting*: sequential training where weights are increased for badly predicted instances so far

Algorithmic level

Feature subsets

- when features are randomly selected repeatedly at different stages of the algorithm
- *Random Forests*

Natural instability

- complex models with strong dependence to the random initialization
- typical example: *Neural networks*
- could be forced further:
  - ensemble training
  - penalty for correlated prediction
  - *Negative Correlation Learning*

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Bagging

Boostrap Aggregating

- Breiman (1996)
- simple and efficient
- consists simply in training models on bootstrap samples!
- adapted to models with high variance such as decision trees

Bagging and imbalanced data

- a bootstrap sample contains roughly 63.2% of the original data: could induce serious balance issues
- stratified bootstrap must be used
- avoid balancing the data *before* the bagging

Out-of-bag estimate

Principle

- leverage the fact that a bootstrap sample contains only 63.2% of the original data
- compute the prediction for a data point x using only the models for which x was not in the bootstrap sample

Formal definition

- K bootstrap samples $\mathcal{D}_1, \ldots, \mathcal{D}_K$ with associated models $g_1, \ldots, g_K$
- O_i: the set of indices k for each $\mathbf{X}_i \notin \mathcal{D}_k$
- risk estimate (averaging case):

$$\widehat{R}_\ell^{oob}(g_{bag}) = \frac{1}{N} \sum_{i=1}^{N} \ell \left(\frac{1}{|O_i|} \sum_{k \in O_i} g_k(\mathbf{X}_i), \mathbf{Y}_i \right)$$

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Increasing the diversity

Feature subsets

- the bootstrap might not induce enough diversity
- adding random subspace might also be insufficient

Random Forest

- A Random Forest is an ensemble of Decision Trees, generally trained via the bagging method
- Local random feature subsets
- The Random Forest algorithm introduces extra randomness when growing trees:
  - Normal Decision Trees (standard CART approach): chose the best feature among all the features
  - Random Forests: chose the best feature among a random subset of the features
    - This results in a greater tree diversity, which trades a higher bias for a lower variance, generally yielding an overall better model.
- no pruning

Random Forests

In practice

- one of the best state-of-the-art solution for classical data
- leverage the out-of-bag estimate (all in one solution)
- parallel computation of the trees
- classical parameters:
  - at least 500 trees (a.k.a. bootstrap samples), depends on the complexity of the data
  - the "optimal" size of the subset of variables depends on several aspects:
    - highly correlated variables do not need a large subset (!)
    - Breiman recommended to use $P/3$ for regression problems and $\sqrt{P}$ for classification problems
  - the parameters can (should) be optimized with cross-validation

Extra-Trees

- It is possible to make trees even more random by also using random thresholds for each feature.
- A forest of such extremely random trees is simply called an Extremely Randomized Trees ensemble (Extra-Trees).
  - Extra-Trees are much faster to train than regular Random Forests since finding the best possible threshold for each feature at every node is one of the most time-consuming tasks of growing a tree.

Feature Importance (XAI)

- Random Forests are very handy if you need to perform feature selection.
- Important features are likely to appear closer to the root of the tree. Hence, to get an estimate of a feature's importance average depth at which it appears across all trees in the forest.
  - In the Iris dataset, the most important features are the petal length (44%) and width (42%), while sepal length and width are at 11% and 2%.

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Boosting

- Boosting (originally called *hypothesis boosting*) refers to any Ensemble method that can combine several weak learners into a strong learner.
- The idea of most boosting methods is to train predictors sequentially, each trying to correct its predecessor.
- There are many boosting methods available, but by far the most popular are *AdaBoost* (short for *Adaptive Boosting*) and *Gradient Boosting*.

Boosting

Principle

- Freund and Schapire (AdaBoost 1997)
- sequential learning of an ensemble of models
- weight based variability: each model is obtained from the same data set but with different weights
- weights emphasize badly predicted examples

Original AdaBoost

- classification $\mathcal{Y} = \{-1, 1\}$ and $\ell(p, t) = \mathbb{1} p \neq t$
- binary or score based models $g_k(\mathbf{x}) = \text{sign}(f_k(\mathbf{x}))$

AdaBoost

set $w_{1,i} = \frac{1}{N}$ for all i
for $k = 1$ to K **do**
    learn g_k on $(\mathbf{X}_i, \mathbf{Y}_i)_{1 \leq i \leq N}$ using the weights $w_{k,i}$
    compute $\epsilon_k = \sum_i w_{k,i} \mathbb{1} g_k(\mathbf{X}_i) \neq \mathbf{Y}_i$
    compute $\alpha_k = \frac{1}{2} \log \frac{1-\epsilon_k}{\epsilon_k}$
    compute $w_{k+1,i} = w_{k,i} \exp(-\alpha_k \mathbf{Y}_i g_k(\mathbf{X}_i))$
    normalize the weights $w_{k+1,i}$
end for
final model $g = \text{sign}\left(\sum_k \alpha_k g_k\right)$

Interpretation

Analysis

- $\epsilon_k = \sum_i w_{ki} \mathbb{1} g_k(\mathbf{X}_i) \neq \mathbf{Y}_i$: weighted empirical risk
- $\exp(-\alpha_k \mathbf{Y}_i g_k(\mathbf{X}_i))$:
  - if $\mathbf{Y}_i = g_k(\mathbf{X}_i)$, the prediction is correct and the weight is divided by $\exp(\alpha_k)$
  - if $\mathbf{Y}_i \neq g_k(\mathbf{X}_i)$, the prediction is incorrect and the weight is multiplied by $\exp(\alpha_k)$
- $\exp(\alpha_k) = \sqrt{\frac{1-\epsilon_k}{\epsilon_k}}$:
  - no effect when g_k is a bad model (i.e. when ϵ_k is close to $\frac{1}{2}$)
  - strong effect when g_k is a good model

Additive models

Model

- an additive model has the form

$$g(\mathbf{x}) = \sum_{k=1}^{K} \beta_k f(\mathbf{x}, \gamma_k),$$

  where $f(., \gamma_k)$ is a general parametric model (e.g. a tree)
- classical risk minimisation is complex (doable with respect to β_k complex in general with respect to γ_k)

Sequential optimization

- $h_k(\mathbf{x}) = \sum_{m=1}^{k} \beta_m f(\mathbf{x}, \gamma_m)$ $(g = h_K)$
- $h_k(\mathbf{x}) = h_{k-1}(\mathbf{x}) + \beta_k f(\mathbf{x}, \gamma_k)$
- optimize h_1, fix its parameters, then optimize h_2 with respect to β_2 and γ_2 only, etc.

Exponential loss

Standard surrogate loss

- $\ell_e(g(\mathbf{x}), \mathbf{y}) = e^{-\mathbf{y}g(\mathbf{x})}$
- upper convex approximation of the binary loss

Sequential optimization

- additive to multiplicative

$$\begin{aligned}\ell_e(h_k(\mathbf{x}), \mathbf{y}) &= e^{-\mathbf{y}h_{k-1}(\mathbf{x})} e^{-\beta_k \mathbf{y} f(\mathbf{x}, \gamma_k)} \\ &= \ell_e(h_{k-1}(\mathbf{x}), \mathbf{y}) e^{-\beta_k \mathbf{y} f(\mathbf{x}, \gamma_k)}\end{aligned}$$

- empirical loss, with $w_{k,i} = \frac{1}{N}\ell_e(h_{k-1}(\mathbf{X}_i), \mathbf{Y}_i)$ (and $w_{1,i} = \frac{1}{N}$):

$$\widehat{R}_{\ell_e}(h_k) = \sum_{i=1}^{N} w_{k,i} e^{-\beta_k \mathbf{Y}_i f(\mathbf{X}_i, \gamma_k)}$$

Link with AdaBoost

Binary models

- $f(\mathbf{x}, \gamma_k) \in \{-1, 1\}$
- therefore

$$e^{-\beta_k \mathbf{Y}_i f(\mathbf{X}_i, \gamma_k)} = \begin{cases} e^{-\beta_k} & \text{if } f(\mathbf{X}_i, \gamma_k) = \mathbf{Y}_i \\ e^{\beta_k} & \text{if } f(\mathbf{X}_i, \gamma_k) \neq \mathbf{Y}_i \end{cases}$$

- empirical loss

$$\begin{aligned} \widehat{R}_{\ell_e}(h_k) &= e^{-\beta_k} \sum_{f(\mathbf{X}_i, \gamma_k) = \mathbf{Y}_i}^{N} w_{k,i} + e^{\beta_k} \sum_{f(\mathbf{X}_i, \gamma_k) \neq \mathbf{Y}_i}^{N} w_{k,i} \\ &= (e^{\beta_k} - e^{-\beta_k}) \sum_{i=1}^{N} w_{k,i} \ell_b(f(\mathbf{X}_i, \gamma_k), \mathbf{Y}_i) + e^{-\beta_k} \sum_{i=1}^{N} w_{k,i} \end{aligned}$$

where ℓ_b is the binary loss

Link with AdaBoost

Solving for γ_k

- both the weights $w_{k,i}$ and β_k are held constant
- then we have a standard (weighted) empirical risk minimization:

$$\arg\min_{\gamma_k} \widehat{R}_{\ell_e}(h_k) = \arg\min_{\gamma_k} \frac{1}{N} \sum_{i=1}^{N} w_{k,i} \ell_b(f(\mathbf{X}_i, \gamma_k), \mathbf{Y}_i)$$

Solving for β_k

- both the weights $w_{k,i}$ and γ_k are held constant
- if we assume $\sum_i w_{k,i} = 1$, then

$$\arg\min_{\beta_k} \widehat{R}_{\ell_e}(h_k) = \frac{1}{2} \log \frac{1 - \epsilon_k}{\epsilon_k} = \alpha_k,$$

with $\epsilon_k = \sum_{i=1}^{N} w_{k,i} \ell_b(f(\mathbf{X}_i, \gamma_k), \mathbf{Y}_i)$

Link with AdaBoost

Weights

- $w_{1,i} = \frac{1}{N}$: initial set
- update

$$\begin{aligned} w_{k+1,i} &= \ell_e(h_k(\mathbf{X}_i), \mathbf{Y}_i), \\ &= w_{k,i} e^{-\beta_k \mathbf{Y}_i f(\mathbf{X}_i, \gamma_k)}, \\ &= w_{k,i} e^{-\alpha_k \mathbf{Y}_i f(\mathbf{X}_i, \gamma_k)}, \end{aligned}$$

using the optimal value for β_k

In summary

- AdaBoost is a form of sequential learning for $g(\mathbf{x}) = \sum_{k=1}^{K} \beta_k f(\mathbf{x}, \gamma_k)$ with the exponential loss
- with independent optimisation of β_k and γ_k

Gradient Boosting

Generalization

- sequential learning of additive models
- arbitrary loss function

Steepest descent

- iterative minimization of $\widehat{R}_\ell(g)$:
  1. start with some initial model g_0
  2. update the model following the steepest descent: $g_{k+1} = g_k - \rho \nabla \widehat{R}_\ell(g_k)$
- leads to an additive model if $\nabla \widehat{R}_\ell(g_k)$ is a function of the form $f(., \gamma_k)$
- *gradient boosting*: approximate $\nabla \widehat{R}_\ell(g_k)$ with $f(., \gamma_k)$

Gradient Boosting

In the following algorithm, ℓ' is the derivative of the loss function ℓ with respect to the prediction

set g_0 to the best constant function: $g_0(\mathbf{x}) = \arg\min_{\mathbf{y}} \sum_i \ell(\mathbf{y}, \mathbf{Y}_i)$
for $k = 1$ to K **do**
  compute $r_{k,i} = -\ell'(g_{k-1}(\mathbf{X}_i), \mathbf{Y}_i)$
  learn f_k on $(\mathbf{X}_i, r_{k,i})_{1 \leq i \leq N}$
  compute

$$\rho_k = \arg\min_{\rho} \sum_i \ell(g_{k-1}(\mathbf{X}_i) + \rho f_k(\mathbf{X}_i), \mathbf{Y}_i)$$

  set $g_k = g_{k-1} + \rho_k f_k$
end for
final model g_K

Regularization

The case of trees

- one could use arbitrary trees for f_k
- tends to overfit and to slow down learning
- common practice: limit the complexity of the trees
- typical choices: between 4 and 10 leaves

General regularization

- size of the ensemble: K
- shrinkage: replace $g_k = g_{k-1} + \rho_k f_k$ by $g_k = g_{k-1} + \nu \rho_k f_k$ (with $0 < \nu < 1$)
- standard regularization terms (e.g. ridge like)

Boosting

In practice

- one of the best state-of-the-art solution for classical data
- very fast implementations such as XGBoost
- numerous practical tricks (e.g. including local random feature subsets in tree growing)
- parameters:
  - main one: size of the ensemble (K), must be tuned
  - other parameters can be optimized (e.g., number of leaves) but with less impact on the performances

In summary

Ensemble methods

- Random Forests
- Boosting
- pros and cons
  - \+ state of the art performances
  - \+ straightforward parallel implementation
  - \+ efficient large scale implementations
  - \+ adapted to mixed data
  - \+ handle missing data
  - \- black box models
  - \- high runtime compared to many other models
  - \- high storage cost compared to many other models